Lecture 8

Differentiation rules

Uses.

Summary 2: The exponential, logarithmic and restricted trigonometric functions have their standing school identities and domains (E). Hyperbolic sine and cosine are combinations of exponentials.
Summary 5: Limits obey arithmetic laws. Continuity allows composition when inner values equal the outer base point.
Summary 6: The quotients $\sin h/h$, $(e^h - 1)/h$ and $\ln(1+h)/h$ tend to 1. The cosine quotient satisfies $(1 - \cos h)/h^2 \to 1/2$.
Summary 7: A derivative is a finite difference-quotient limit. Differentiability implies continuity and gives a linear approximation. A continuous strictly monotone interval function has a continuous inverse.

Objectives. *The derivative definition gives rules for building derivatives from simpler ones. We prove the arithmetic, chain and inverse rules, keeping the conditions needed for division and composition. The standard limits then give an elementary derivative table. Every formula is used on a domain where its hypotheses hold.*

By the end of this lecture you will be able to:

- Prove and apply the arithmetic and chain rules for differentiable functions.
- Derive elementary derivatives from the standard limits.
- Differentiate the inverse of a continuous strictly monotone interval function where the original derivative is nonzero.
- Distinguish differentiability from continuity of the derivative.

8.1 Arithmetic rules

Theorem 8.1 (Sum, scalar, product and quotient rules). *Let f and g be real functions on an open interval containing c, both differentiable at c, and let $\lambda \in \mathbb{R}$. Then*

$$(f+g)'(c) = f'(c) + g'(c), \qquad (\lambda f)'(c) = \lambda f'(c),$$

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c).$$

If $g(c) \neq 0$, then f/g is defined on a neighbourhood of c, is differentiable at c, and

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{g(c)^2}.$$

Proof. The difference quotient for $f+g$ is the sum of their difference quotients. Taking limits proves the sum rule. Multiplying a quotient by λ proves the scalar rule. For the product, add and subtract $f(c)g(c+h)$ in its numerator:

$$\frac{f(c+h)g(c+h) - f(c)g(c)}{h} = \frac{f(c+h) - f(c)}{h} g(c+h) + f(c)\frac{g(c+h) - g(c)}{h}.$$

Differentiability gives $g(c+h) \to g(c)$. The product and sum limit laws now give the asserted product derivative.

For the quotient, continuity of g and $g(c) \neq 0$ give a radius where $|g(c+h)| > |g(c)|/2$. Thus all the following divisions are valid there. Combining fractions and splitting the numerator gives

$$\frac{f(c+h)/g(c+h) - f(c)/g(c)}{h} = \frac{g(c)\,[f(c+h) - f(c)]/h - f(c)\,[g(c+h) - g(c)]/h}{g(c+h)g(c)}.$$

The denominator tends to $g(c)^2 \neq 0$. The arithmetic limit laws prove the stated quotient rule. ∎

For example, applying the product rule to $x \cdot x$ gives $2x$, as the direct difference quotient did in Lecture 7. More generally, induction gives the positive-integer power rule

$$(x^n)' = nx^{n-1} \qquad (n \in \mathbb{N},\ x \in \mathbb{R}). \tag{8.1}$$

The base case $n = 1$ is the identity derivative 1. If the rule holds at n, the product rule gives $(x^{n+1})' = nx^{n-1}x + x^n = (n+1)x^n$. Here x^0 means the constant polynomial 1, including at zero. Constants have derivative zero, so finite sums give the derivative of every polynomial. The quotient rule then gives derivatives of rational functions at inputs where their denominators are nonzero. For instance, on $\mathbb{R}$,

$$\left(\frac{x^2}{1+x^2}\right)' = \frac{2x(1+x^2) - 2x^3}{(1+x^2)^2} = \frac{2x}{(1+x^2)^2}.$$

The denominator is strictly positive, so the derivative exists everywhere.

8.2 The chain rule

Theorem 8.2 (Chain rule). *Let $f : I \to \mathbb{R}$ be differentiable at c in the open interval I. Let $g : J \to \mathbb{R}$ be differentiable at $u = f(c)$ in the open interval J, and suppose $f(x) \in J$ for all x sufficiently near c. Then $g \circ f$ is differentiable at c, with*

$$(g \circ f)'(c) = g'(f(c))f'(c).$$

Proof. The linear approximation criterion for g gives

$$g(u+k) - g(u) = g'(u)k + r(k), \qquad r(k) = o(k), \quad r(0) = 0.$$

To handle a zero inner change, extend the quotient of this remainder by its increment. Define $\widetilde{r}(k) = r(k)/k$ for $k \neq 0$ and $\widetilde{r}(0) = 0$. The little-oh condition gives $\widetilde{r}(k) \to 0$, so this extension is continuous at zero. The identity becomes

$$g(u+k) - g(u) = \big(g'(u) + \widetilde{r}(k)\big)k,$$

and remains valid at $k = 0$. Put $k = f(c+h) - f(c)$. Differentiability of f implies continuity, so $k \to 0$. For nonzero h sufficiently near zero,

$$\frac{g(f(c+h)) - g(f(c))}{h} = \big(g'(u) + \widetilde{r}(f(c+h) - f(c))\big)\frac{f(c+h) - f(c)}{h}.$$

Continuity of $\widetilde{r}$ at zero makes the first factor tend to $g'(u)$. The second tends to $f'(c)$. The product law proves the formula. This proof includes increments where $f(c+h) = f(c)$. Dividing by that inner change instead would lose those inputs. ∎

For example, take $F(x) = (1+x^2)^3$ on $\mathbb{R}$. The inner function has derivative $2x$ and the outer cubic has derivative $3u^2$. The chain rule gives

$$F'(x) = 3(1+x^2)^2 \cdot 2x = 6x(1+x^2)^2.$$

The outer derivative is evaluated at the inner value before the two factors are multiplied. The differential notation keeps the same structure: if $u = f(x)$ and $y = g(u)$, then $du = f'(x)\,dx$ and $dy = g'(u)\,du$ give $dy = g'(f(x))f'(x)\,dx$. These are linear changes at the fixed base points. Their form follows from the chain rule, rather than from cancellation of symbols treated as numerical fractions.

8.3 Elementary derivatives from standard limits

For $f(x) = e^x$, the exponential identity gives

$$\frac{e^{c+h} - e^c}{h} = e^c \frac{e^h - 1}{h} \longrightarrow e^c.$$

Thus $(e^x)' = e^x$ on $\mathbb{R}$. For $f(x) = \ln x$ and $c > 0$, keep $|h| < c$ to stay in its domain. The logarithm identity gives

$$\frac{\ln(c+h) - \ln c}{h} = \frac{1}{c}\frac{\ln(1+h/c)}{h/c} \longrightarrow \frac{1}{c}.$$

The inner argument is nonzero when $h \neq 0$. Thus $(\ln x)' = 1/x$ on $(0, \infty)$.

For sine, the addition identity gives

$$\frac{\sin(c+h)-\sin c}{h}=\sin c\frac{\cos h-1}{h}+\cos c\frac{\sin h}{h}.$$

The first quotient tends to zero because $(\cos h-1)/h=-h(1-\cos h)/h^2\to 0$. The second tends to 1. Thus $(\sin x)'=\cos x$ on $\mathbb{R}$. The cosine addition identity similarly gives

$$\frac{\cos(c+h)-\cos c}{h}=\cos c\frac{\cos h-1}{h}-\sin c\frac{\sin h}{h}\longrightarrow -\sin c.$$

This proves $(\cos x)'=-\sin x$ on $\mathbb{R}$.

The remaining direct elementary formulas now follow from proved rules. For $a>0$, the identity $a^x=e^{x\ln a}$ gives $(a^x)'=a^x\ln a$, including the constant case $a=1$. For $x>0$ and $\alpha\in\mathbb{R}$, the identity $x^\alpha=e^{\alpha\ln x}$ gives

$$(x^\alpha)'=e^{\alpha\ln x}\frac{\alpha}{x}=\alpha x^{\alpha-1}.$$

This formula on positive inputs does not assign a real power to a negative base. Integer powers already have their larger domains by Eq. (8.1) and the quotient rule. For $a>0$, $a\neq 1$, defining $\log_a x=\ln x/\ln a$ gives $(\log_a x)'=1/(x\ln a)$ for $x>0$.

The tangent is $\sin x/\cos x$. Its derivative is

$$(\tan x)'=\frac{\cos^2 x+\sin^2 x}{\cos^2 x}=\frac{1}{\cos^2 x}\qquad(\cos x\neq 0).$$

Defining $\cot x=\cos x/\sin x$ similarly gives $(\cot x)'=-1/\sin^2 x$ when $\sin x\neq 0$. These formulas are local to the intervals avoiding their denominator zeros.

Variable exponents can be differentiated through logarithms as well. If a and b are differentiable on an open interval and $a(x)>0$ there, then $Y(x)=a(x)^{b(x)}=\exp(b(x)\ln a(x))$ is differentiable. The chain and product rules give

$$Y'(x)=a(x)^{b(x)}\left(b'(x)\ln a(x)+b(x)\frac{a'(x)}{a(x)}\right).\tag{8.2}$$

The same formula is obtained by differentiating $\ln Y=b\ln a$, giving $Y'/Y=b'\ln a+ba'/a$. This procedure is called *logarithmic differentiation*. The exponential representation first justifies differentiability, so taking the derivative of $\ln Y$ does not assume an unproved derivative of Y. For example, on $x>0$,

$$\left(x^{\sin x}\right)'=x^{\sin x}\left(\cos x\ln x+\frac{\sin x}{x}\right).$$

The ordinary constant-exponent power rule would omit the first term. Both a varying base and a varying exponent contribute to this derivative.

For example, let $H(x)=\ln(1+x^2)/(1+x^2)$. Both logarithm and denominator are defined for every $x\in\mathbb{R}$, since $1+x^2>0$. The chain and quotient rules give

$$H'(x)=\frac{[2x/(1+x^2)](1+x^2)-2x\ln(1+x^2)}{(1+x^2)^2}=\frac{2x[1-\ln(1+x^2)]}{(1+x^2)^2}.$$

For $K(x)=xe^{-x^2}$, the product and chain rules instead give $K'(x)=e^{-x^2}(1-2x^2)$ on $\mathbb{R}$.

8.4 The derivative of an inverse

Theorem 8.3 (Inverse derivative). *Let $f : I \to \mathbb{R}$ be continuous and strictly monotone on an open interval I. Suppose f is differentiable at $c \in I$ and $f'(c) \neq 0$. Write $g : f(I) \to I$ for its inverse. Then g is differentiable at $v = f(c)$ and*

$$g'(v) = \frac{1}{f'(c)} = \frac{1}{f'(g(v))}.$$

Proof. The image is an interval, and v is an interior point of it: choose points of I on both sides of c and use their strictly ordered images to bracket v. Continuity of the inverse, proved in Lecture 7, gives $g(v+k) \to c$ as $k \to 0$ within the image. For $k \neq 0$, put $x = g(v+k)$. Injectivity gives $x \neq c$, and $f(x) - f(c) = k$. Thus

$$\frac{g(v+k) - g(v)}{k} = \frac{x - c}{f(x) - f(c)} = \left(\frac{f(x) - f(c)}{x - c}\right)^{-1} \longrightarrow \frac{1}{f'(c)}.$$

The reciprocal limit is valid because $f'(c) \neq 0$. This proves differentiability of g and the formula. ∎

Apply this theorem to sine on $(-\pi/2, \pi/2)$. For $u = \arcsin x$ with $-1 < x < 1$, its cosine is positive and satisfies $\cos u = \sqrt{1 - x^2}$. Therefore

$$(\arcsin x)' = \frac{1}{\sqrt{1 - x^2}} \qquad (-1 < x < 1).$$

For cosine on $(0, \pi)$, its derivative is $-\sin u$ and its sine is positive. The same theorem gives $(\arccos x)' = -1/\sqrt{1 - x^2}$ for $-1 < x < 1$. For tangent on $(-\pi/2, \pi/2)$, its derivative is $1 + \tan^2 u > 0$, so $(\arctan x)' = 1/(1 + x^2)$ for every real x. The inverse sine and cosine remain defined at their domain endpoints, but the inverse-derivative proof above does not apply there.

These proved formulas give the following lookup table. Each derivative is asserted on the domain printed beside it.

Function	Derivative	Domain for the formula
x^α	$\alpha x^{\alpha-1}$	$x > 0$, $\alpha \in \mathbb{R}$
e^x	e^x	$\mathbb{R}$
a^x	$a^x \ln a$	$\mathbb{R}$, $a > 0$
$\ln x$	$1/x$	$x > 0$
$\sin x$, $\cos x$	$\cos x$, $-\sin x$	$\mathbb{R}$
$\tan x$	$1/\cos^2 x$	$\cos x \neq 0$
$\arcsin x$, $\arccos x$	$1/\sqrt{1 - x^2}$, $-1/\sqrt{1 - x^2}$	$-1 < x < 1$
$\arctan x$	$1/(1 + x^2)$	$\mathbb{R}$

8.5 Continuity of the derivative

The existence of a derivative does not imply that the derivative function is continuous. Define $f : \mathbb{R} \to \mathbb{R}$ by $f(0) = 0$ and $f(x) = x^2 \sin(1/x)$ for $x \neq 0$. At zero the difference quotient is $h \sin(1/h)$, whose absolute value is at most $|h|$. Squeeze gives $f'(0) = 0$. For nonzero x, the proved rules give

$$f'(x) = 2x \sin(1/x) - \cos(1/x).$$

Along $x_n = 1/(2\pi n)$ these derivatives equal -1. Along $y_n = 1/((2n+1)\pi)$ they equal 1. Both input sequences tend to zero, so f' is not continuous at zero, although f is differentiable everywhere. A differentiable function on an open interval I whose derivative is continuous on I is called *continuously differentiable*. The class of these functions is denoted $C^1(I)$. The example is differentiable but is not C^1 on any interval containing zero.

8.6 Hyperbolic functions

From the exponential definitions in Lecture 2 and the chain rule,

$$(\sinh x)' = \frac{e^x + e^{-x}}{2} = \cosh x, \qquad (\cosh x)' = \frac{e^x - e^{-x}}{2} = \sinh x.$$

The identities also give $\cosh^2 x - \sinh^2 x = 1$ by expanding the squares. For $z = e^x > 0$, the inequality $(z-1)^2 \geq 0$ gives $z + 1/z \geq 2$. Thus $\cosh x \geq 1$, so division by it is always allowed. Writing $\tanh x = \sinh x/\cosh x$ and $\operatorname{sech} x = 1/\cosh x$ as on Sheet 2, the quotient rule gives

$$(\tanh x)' = \frac{1}{\cosh^2 x}, \qquad (\operatorname{sech} x)' = -\frac{\sinh x}{\cosh^2 x} = -\operatorname{sech} x \, \tanh x.$$

All four derivative formulas hold on $\mathbb{R}$.

Sheet 2 derives the logarithmic expressions for the following real branches. Here the domains are domains of the functions; the derivative formulas below may use smaller open intervals at endpoints.

inverse branch	expression	domain
$\operatorname{arsinh} x$	$\ln(x + \sqrt{1+x^2})$	$x \in \mathbb{R}$
$\operatorname{arcosh} x$	$\ln(x + \sqrt{x^2-1})$	$x \geq 1$
$\operatorname{artanh} x$	$\frac{1}{2}\ln\big((1+x)/(1-x)\big)$	$-1 < x < 1$
$\operatorname{arsech} x$	$\operatorname{arcosh}(1/x)$	$0 < x \leq 1$

The original cosh and sech are restricted to $[0, \infty)$ to make these inverse branches single-valued. Their derivatives follow directly from the rules already proved. For $r = \sqrt{1+x^2}$ we have $r > |x|$, so $x + r > 0$, and $r' = x/r$. Consequently

$$\big(\operatorname{arsinh} x\big)' = \big(\ln(x+r)\big)' = \frac{1 + x/r}{x+r} = \frac{1}{\sqrt{1+x^2}} \qquad (x \in \mathbb{R}).$$

For $x > 1$, put instead $r = \sqrt{x^2-1} > 0$. Again $r' = x/r$, and the same logarithm calculation gives $(\operatorname{arcosh} x)' = 1/\sqrt{x^2-1}$. This formula does not include the domain endpoint $x = 1$. For $-1 < x < 1$, both $1+x$ and $1-x$ are positive, and

$$\big(\operatorname{artanh} x\big)' = \frac{1}{2}\left(\frac{1}{1+x} + \frac{1}{1-x}\right) = \frac{1}{1-x^2}.$$

Finally, Sheet 2 gives $\operatorname{arsech} x = \operatorname{arcosh}(1/x)$ on $0 < x \leq 1$. For $0 < x < 1$, the chain rule gives

$$\big(\operatorname{arsech} x\big)' = \frac{-1/x^2}{\sqrt{1/x^2 - 1}} = -\frac{1}{x\sqrt{1-x^2}}.$$

The final simplification uses $x > 0$ to replace $\sqrt{(1-x^2)/x^2}$ by $\sqrt{1-x^2}/x$. All four formulas retain the real branches specified on Sheet 2.

Summary 8.

- Theorem 8.1: sums and products of differentiable functions are differentiable. A quotient also requires a nonzero denominator.
- Theorem 8.2: $(g \circ f)'(c) = g'(f(c))f'(c)$ when both pointwise derivatives exist and the composition is defined near c.
- Standard limits prove the exponential, logarithmic, sine and cosine derivatives. The arithmetic and chain rules give the power and tangent formulas.
- Theorem 8.3: a continuous strictly monotone interval function has inverse derivative $1/f'(c)$ at the image of c if $f'(c) \neq 0$. This gives the inverse trigonometric derivatives on their stated interiors.
- Hyperbolic derivatives follow from their exponential definitions. Differentiability alone does not imply continuity of the derivative.

Lecture 9 applies these rules to given implicit branches and parametrised curves, then proves the mean value theorems.

References

R. G. Bartle and D. R. Sherbert, *Introduction to Real Analysis*, 4th ed., Wiley, 2011, Chapter 6.